
CLINTON ENWEREM

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PROFESSIONAL SUMMARY

Ph.D. Candidate and Robotics Research Engineer with **4+ years** building multifingered manipulation systems that stay reliable when perception and contact are uncertain.

- Dexterous grasping and manipulation under multimodal uncertainty, spanning risk-sensitive grasp synthesis, belief-space estimation, safety-critical control, safe reinforcement learning, and behavior cloning, validated on real hardware (Fairino FR3, xArm7, LEAP Hand v1, RealHand L6).
- Research-grade ROS 2 robot software in Python and C++, covering grasp synthesis and execution libraries, QP-based safety filters, and simulation benchmarking in MuJoCo and Drake.

EDUCATION

University of Maryland, College Park

Ph.D., Electrical & Computer Engineering (Dexterous Manipulation). Expected Fall 2026.

MD, USA

Aug. 2021 – Present

University of Nigeria, Nsukka

B.Eng., Electrical Engineering (Control Theory Emphasis). Highest Honors. GPA: 3.84/4.0 (WES U.S. equivalent).

Enugu, Nigeria

Aug. 2018

WORK EXPERIENCE

Graduate Research Assistant

Institute for Systems Research (ISR), University of Maryland

Aug. 2021 – Present

College Park, MD

- Develop and maintain Python and C++ software for grasp planning, stochastic motion planning, and closed-loop manipulation control on real manipulators (FR3, xArm7, LEAP Hand, RealHand L6).
- Build expert data collection pipelines and train MLP, LSTM, and DP3 behavior cloning policies for grasp stabilization under partial tactile observability.

Robotics Trainee

Robotics & Artificial Intelligence Nigeria

Mar. 2020 – Feb. 2021

Ibadan, Nigeria

- Built ROS-based visual-SLAM, state-estimation, and flight-control software for a low-cost UAV on embedded hardware.

RESEARCH SOFTWARE

GDF, Planner-Free Certified Grasping (Project Page) | Python, MuJoCo, CasADi

2026

46/50 objects grasped and lifted · **94%** median quality margin retained · **0.09 ms** QP solve per 20 ms step · **demonstrated on a Unitree G1 humanoid** and a fixed-base arm. *Stationary feedback law descending a softmin grasp distance field under a CBF-CLF safety filter, with no planner, stored trajectory, or grasp selection step.*

EquiDexFlow, SE(3)-Equivariant Grasp Generation (Project Page, Code) | Python, PyTorch

Jun. 2026

Zero friction violations · **0.46 Nm** wrench residual (best of 4) · **8,100** certified grasps over 81 objects · **6/6** objects on a physical LEAP Hand. *Flow-matching model generating wrist pose, joint angles, contacts, and forces from a point cloud, with friction-cone compliance by construction.*

VNB-Grasp, Variational Neural Belief Parameterizations (Code) | Python, PyTorch, MuJoCo

Apr. 2026

10× faster belief propagation · **0.14** mean absolute calibration error · **82%** grasp success with zero contact violations against 55% for a risk-neutral baseline. *Differentiable Gaussian-mixture contact belief with pathwise CVaR gradients, plus a MuJoCo friction-sensitivity benchmark.*

TECHNICAL SKILLS

Robotics: ROS/ROS2, MuJoCo, Drake, Isaac Sim, Isaac Lab, Pinocchio, MoveIt2; **Control & Safe Learning:** Control Barrier Functions, Safety Filters, Safe & Constrained Reinforcement Learning, Model Predictive Control (MPC), QP-based Control, Belief-Space Planning, Sim-to-Real Transfer; **ML / Perception:** PyTorch, TensorFlow, OpenCV; **Programming:** Python, C++, MATLAB, Bash; **Optimization:** CasADi, CVXPY, qpsolvers, IPOPT, Gurobi; **Dev Tools:** git, Docker; **Robots & EoATs:** Fairino FR3, xArm7, Unitree G1 (simulation), LEAP Hand v1, RealHand L6.

SELECTED PUBLICATIONS

- **C. Enwerem**, J. S. Baras, C. Belta. “Grasp Execution Without a Planner: Configuration-Space Grasp Distance Fields with Certified Safety & Guaranteed Quality.” *Manuscript*, 2026. Project page.
- **C. Enwerem**, J. S. Baras, C. Belta. “FIRMGrasp: A Friction-Informed Risk Margin for Robust Grasp Synthesis.” *Preprint*, 2026. [arXiv:2607.25049](https://arxiv.org/abs/2607.25049).
- **C. Enwerem**, J. S. Baras, C. Belta. “EquiDexFlow: Contact-Grounded SE(3)-Equivariant Dexterous Grasp Generative Flows.” *Under review*, 2026. [arXiv:2606.12728](https://arxiv.org/abs/2606.12728).
- Also to appear 2026 at IROS and CDC, plus 5 peer-reviewed first-author papers (CoDIT’23, L-CSS’24, ECC’24, CDC’24, CDC’25).